

B1.2 What happens in cells (and what do cells need)?

Summary

Life processes depend on biological molecules whose structure is related to their function. Inside every cell is genetic material and this is used as a code to make proteins. Enzymes are important proteins in biology.

Underlying knowledge and understanding

Learners should have a simple understanding of the double helix model of DNA. Learners should be familiar with the idea of enzymes as biological catalysts.

Common misconceptions

Learners commonly hold the misconception that DNA is made of protein or sugar. Learners also think that all enzymes have an optimum temperature of 37°C (human body temperature). The range of optimum temperatures of enzymes should be introduced through the teaching of this topic and further addressed when considering homeostatic mechanisms for controlling temperature.

Tiering

Statements shown in **bold** type will only be tested in the Higher Tier papers. All other statements will be assessed in both Foundation and Higher Tier papers.

Reference	Mathematical learning outcomes	Mathematical skills
BM1.2i	carry out rate calculations for chemical reactions	M1a and M1c
BM1.2ii	understand and use simple compound measures such as the rate of a reaction	M1a and M1c

Topic content		Opportunities to cover:		Practical suggestions
Learning outcomes	To include	Maths	Working scientifically	
B1.2a describe DNA as a polymer			WS1.4a	Production of 3D models of DNA to illustrate its structure.
B1.2b describe DNA as being made up of two strands forming a double helix				
B1.2c ☑ describe that DNA is made from four different nucleotides; each nucleotide consisting of a common sugar and phosphate group with one of four different bases attached to the sugar	the pairs of complementary bases (A-T and G-C)		WS1.4a, WS2a, WS2b, WS2c, WS2d	Production of 3D models of DNA to illustrate its structure. Investigation of DNA extraction from a living organism (e.g. kiwi, leek, onion, wheat germ). (PAG B2)
B1.2d ☑ recall a simple description of protein synthesis	the unzipping of the DNA molecule around the gene, copying to mRNA in nucleus (transcription), (translation) of the nucleotide sequence in the cytoplasm, tRNA as the carrier of amino acids			Comparison of transcription and translation to a non-lending library. Use of kinaesthetic activities to demonstrate transcription and translation.
B1.2e ☑ explain simply how the structure of DNA affects the proteins made in protein synthesis	triplet code and its use to determine amino acid order in a protein			

B1.2f	describe experiments that can be used to investigate enzymatic reactions		M1a, M1c, M2g	WS1.1h, WS1.2b, WS1.2c, WS1.2e, WS1.3a, WS1.3b, WS1.3c, WS1.3d, WS1.3e, WS1.3f, WS1.3g, WS2a, WS2b, WS2c, WS2d	Investigations of enzyme activity, including numerical analysis of data and graphical representation of results. (PAG B2, PAG B4, PAG B6)
B1.2g	explain the mechanism of enzyme action	the role of enzymes in metabolism, the role of the active site, enzyme specificity (lock and key hypothesis) and factors affecting the rate of enzyme controlled reactions (pH, temperature, substrate and enzyme concentration)	M1a, M1c, M3d, M4b	WS2a, WS2b, WS2c, WS2d	Investigation into the effect of amylase on a baby rice paste. (PAG B2, PAG B4, PAG B6) Investigation of enzyme controlled reactions. (PAG B2, PAG B4, PAG B6) Work out rate equations using simple algebraic equations. (PAG B4)